

M001 Week 4 — English Worksheet (Beginner)

Topic: Nested Loops · **Course:** Pyramid Problems · **Level:** Beginner · **Time:** about 30 minutes

A **nested loop** is a loop inside another loop. The inner loop builds one side; the outer loop repeats it for every side.

1 · Predict 🧠

Read the steps. Before you imagine the agent doing it, circle one: line · square · tower.

```
repeat 4 times:
  repeat 3 times:
    place block down
    move forward
  turn right
```

What shape does this make on the ground? Circle one: line · square · tower

How many sides does it have?


```
repeat 3 times:
  place block down
  move forward
```

This is only the inner loop, on its own. What does it build? Circle one: one side · a whole square

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The agent should build a square: 4 sides, turning after each side.

```
# clean
repeat 4 times:
  repeat 3 times:
    place block down
    move forward
  turn right
```

```
# buggy
repeat 4 times:
  repeat 3 times:
    place block down
    move forward
  turn right
```

What is wrong? Look at where **turn right** sits.

Pair B — The agent should build a square with a nested loop, not just one line.

```
# clean
repeat 4 times:
  repeat 3 times:
    place block down
    move forward
  turn right
```

```
# buggy
repeat 3 times:
  place block down
  move forward
turn right
```

What is wrong? How much of the square does the buggy code build?

3 · Fill the Gap

The agent should build a square. After each side, it must turn. One line is missing. Fill it in using the word bank.

```
repeat 4 times:
  repeat 3 times:
    place block down
    move forward
    _____
```

Word bank: turn right · move up by 1 · place block down

Write the missing line:

4 · Tell Me What You Built 📷

Now switch to your homework world. Build a **4x4 square floor** using a nested loop. When you finish, come back here.

Send a photo or video of your floor, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My outer loop repeated ... times.

My inner loop did ...

The hardest part was ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) and show your square. Talk like you are teaching a friend. Try to use these words: **nested loop, outer, inner, repeat.**

1. Show what your nested loop built.
2. Read the outer loop and the inner loop out loud.
3. Show one bug you hit and how you fixed it.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.